

Ng Jin Yang
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PROFILE

Cybersecurity Engineer with experience in enterprise access management, red teaming, and security automation at Visa and CSIT. Develops Python and PowerShell automation tools and works across Windows, Linux, IBM z/OS, and HPE NonStop environments.

PROFESSIONAL WORK EXPERIENCE

Visa Worldwide Pte Limited, Singapore **Feb 2026 – Current**

Cyber Security Engineer

- Administer access management for IBM Mainframe z/OS using RACF and zSecure reporting tools.
- Manage access controls for critical Tandem systems using Safeguard and Xygate security suites.
- Built custom applications to automate UAR processes and daily BAU ServiceNow ticketing/incidents.
- Oversee digital certificate lifecycles on z/OS and HPE NonStop for critical financial apps.
- Led User Access Revalidation (UAR) cycles to guarantee adherence to strict cybersecurity standards.

Centre for Strategic Infocomm Technologies, Singapore

Nov 2024 – Feb 2026

Cyber Security Engineer

- Conducted full-scope red team engagements simulating APT tactics, techniques, and procedures aligned with MITRE ATT&CK in Linux & Windows Operating Systems (Win7/Win10).
- Identified and exploited misconfigurations in Active Directory, escalating privileges from domain user to domain admin.
- Developed custom Python/PowerShell scripts to perform reconnaissance and lateral movement
- Build custom solutions to scan entire system for any suspicious files or folders by storing their metadata.
- Performed basic Reverse Engineering using WinDbg, and IDA.

Synapse Pte. Ltd., Singapore

Feb 2024 – Oct 2024

System Analyst (HealthTech Associate Programme)

- Conducted requirements gathering with business stakeholders and translated needs into functional specifications, ensuring alignment with hospital workflows and solution designs.
- Collaborated with analysts and users throughout the SDLC to implement and enhance systems.
- Supported the national project, NGEMR in onboarding organizations such as Woodlands Health to integrate the electronic medical record system to enhance care management and delivery.

INTERNSHIP EXPERIENCE

Shopee Pte. Ltd., Singapore

May 2023 – Aug 2023

Data Analyst Intern

- Efficiently extract data for end-users from several large and complex data warehouses by utilizing SQL
- Automated data collection and aggregation processes to build reporting dashboards on Data Studio
- Optimized billing and operation processes by automating them using Python and relevant libraries such as Pandas, xlswriter, os, googleapiclient, etc., effectively reducing process time and workload

Undercurrent Capital Pte. Ltd., Singapore

May 2022 – Jul 2022

Algorand Specialist, Software Engineering Intern

- Build and test decentralized applications for users to interact using Docker.
- Deployed non-fungible tokens and Algorand currency to smart contracts using Python and JavaScript.
- Learn courses that include Accounts/Wallets, Assets/Transactions, Algorand Virtual Machines, and Algorand Smart Contracts.

SKILLS / INTERESTS

Programming and scripting: Python, PowerShell, Bash, JavaScript, SQL, Java

Security: Red Teaming, Adversary Emulation, Penetration Testing, Access Management, Certificate Lifecycle Management, Threat Modeling, Active Directory

Systems: Linux (Kali, Ubuntu), Windows Server, IBM z/OS, HPE NonStop

Networking & Security: TCP/IP, Firewalls, IDS/IPS, VPN, Network Forensics

Tools: RACF, zSecure, Safeguard, Xygate, Metasploit, Burp Suite, Nmap, Wireshark, WinDbg, IDA

Development: Node.js, Express.js, MongoDB, Docker

External Certifications: Algo Foundry Fundamentals, NotSoSecure Hacking 101

Languages: English, Chinese

Interests: Table Tennis, Gym Training, Running, Building Projects, Piano Music

EDUCATION

Nanyang Technological University, Singapore

Aug 2020 – Dec 2023

Bachelor of Engineering (Computer Science) with a Second Major in Business

- Specialization: Artificial Intelligence & Cyber Security
- Relevant Modules: Database System Principles, Machine Learning, Computer Security, Artificial Intelligence, Advanced Software Engineering, Object Oriented Design & Programming, Data Structure & Algorithms, Algorithm Design & Analysis.

ACADEMIC PROJECTS

Nanyang Technological University, Singapore

Final Year Project – Create Your Own Project

Jan 2023 – Nov 2023

Title: Qualitative Analysis on Students' Online Lecture Notes

- Build a website application to provide a collaborative learning platform by offering them a digital space where they can upload, access, and explore a diverse range of educational materials.
- Implemented the application with MongoDB, Express.js, and Node.js (MEN) Stack.
- Employed advanced data analytical tools, such as Natural Language Processing (NLP) to evaluate the quality and relevance of the uploaded documents, providing insights such as the document quality metrics, and readability scores.
- Conducted qualitative analysis on a sample of courses from various specialisations, and effectively gathered insights on the various metrics on the lecture notes.

Cyber-Physical Systems Security Project

Mar 2023 – Apr 2023

Title: Correlation Power Analysis (CPA)

- Implemented CPA to extract secret information from a cryptographic device by analyzing the power traces and calculating their hamming weight.
- Effectively plot the correlation of all possible key bytes to recover the complete 32-byte key on the Advanced Encrypted Standard (AES) and recover the initialization vector using Python

Multi-Disciplinary Project

Aug 2022 – Oct 2022

Title: Autonomous Image Recognition Task & Fastest Robot Movement Task

- Designed a modified A* path finder algorithm for our robot to visit all 4-sided obstacles at a specific side in the shortest time and distance with a Hamiltonian path using Python.
- Build a simulator with PyGame to test the designed algorithm with the visual movement of the expected paths taken.
- Integrated with Raspberry Pi and the STM32407 hardware to perform the various tasks at hand
- Ranked top 6 in the fastest run time among 47 other teams and captured all 6 images successfully; achieved a fast runtime of 3 minutes.